



Studies on children have found that growing up poor or deprived can impair the development of areas related to memory, language processing, decision-making, and self-control. However, a safe, happy home, with interesting things to do, seems to reduce the harm.



Chronic emotional stress in children can impair development of the amygdala, hippocampus, and frontal lobes, leading to problems with memory, emotion, and learning. It restricts the action of genes regulating the growth of networks of neurons. However, moderate "positive" stress (fun) can aid learning.



A healthy diet (see pp.54–55) rich in omega-3 fatty acids, B vitamins, and antioxidants keeps blood vessels healthy, improving blood flow to the brain. These nutrients have also been linked to improving memory and maintaining cognitive functions in older people.

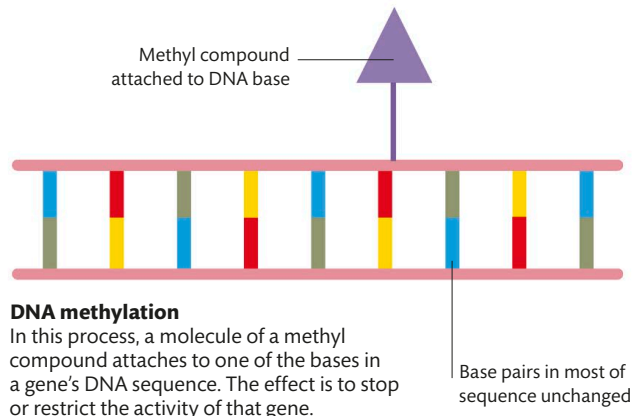


Loneliness has been found to alter the production of neurotransmitters, so people perceive less reward from social contact and are more likely to misinterpret others' attitudes as threatening. However, maintaining close social ties can support memory and cognitive skills.



Epigenetic changes

Changes in the way genes are used (or expressed) that occur during a person's lifetime are called epigenetic changes. They affect gene function, rather than gene structure, and can be passed on to a person's children, although they may last for only a few generations. In the brain, they can influence functions such as learning, memory, reward-seeking, and response to stress. There are two main forms: methylation, in which a compound joins on to the DNA; and histone modification, which alters how tightly the DNA is coiled.

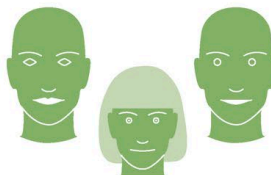


STUDYING TWINS

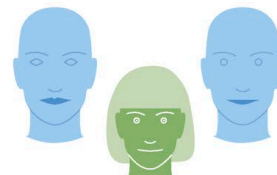
Studies of twins reveal how much of a specific trait, such as intelligence quotient (IQ), is due to inheritance and how much is due to environment. Most twins grow up in the same home; however, identical twins share 100 percent of their genes, while nonidentical (fraternal) twins share only 50 percent. If a trait is more evident in identical twins than in fraternal ones, or appears in identical twins who were separated at birth, it suggests that genetics has a stronger influence than environment.

BIOLOGICAL PARENTS

ADOPTIVE PARENTS



NON-ADOPTED TWIN



ADOPTED TWIN